ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO2 PROBLEM-SOLVING ASSIGNMENT

Standards-Compliant Laser Transmitter Design, Wavelength Control and Nonlinearity Management for an 8-Channel DWDM Link

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Course	Optoelectronics and Optical Communication
Assignment	CO2 Problem-Solving Assignment

Engineering design study for an 8-channel, 25 Gb/s per-channel DWDM transmitter operating near 1550 nm.

Executive Summary

This study designs an 8-channel DWDM transmitter around 1550 nm with 100 GHz channel spacing and 25 Gb/s NRZ per channel. The calculated emission wavelength is 1.550 µm (1550 nm), internal quantum efficiency is 80.0%, and laser output at 40 mA bias is 8.4 mW (9.24 dBm). The required NRZ bandwidth is 17.5 GHz, so the stated 20 GHz bandwidth gives 2.5 GHz of nominal margin. Temperature drift without control is 4 nm, which is five times the 0.8 nm channel spacing. TEC-based temperature control is therefore essential.

For each 80 km span, the effective length is 21.17 km and the estimated SPM phase shift is 0.055 rad per span, or 0.330 rad over six spans under the stated assumption of 2 mW launch power being restored at each span. The supplied SBS estimate is about 1.59 mW, below the 2 mW per-channel launch power, whereas the SRS estimate is about 1.01 W and is far above the per-channel launch power. Soliton order is approximately 0.171, so fundamental soliton operation is not supported by the stated launch power. The recommended architecture is a TEC-stabilized DFB laser array with external modulators, wavelength locking, optical multiplexing, controlled launch power and receiver-side DSP.

Design assumptions

The fiber attenuation coefficient is converted from 0.2 dB/km to 0.04605 km■¹ (power attenuation in nepers per kilometre) before calculating effective length. For the six-span SPM estimate, each amplified span is assumed to launch again at 2 mW per channel; without span-by-span amplification, the power would decrease along the link and the accumulated phase would be lower. The 2 mW launch value is treated as the pulse peak power for the soliton-order estimate. Actual compliance and eye-safety classification require equipment-specific measurements and the applicable standard limits.

Stage 1 – Interpret the Communication Requirement

The link carries eight wavelength channels separated by 100 GHz, with each channel transmitting 25 Gb/s NRZ data. The aggregate line rate is 8 × 25 = 200 Gb/s before framing and overhead. The link contains six 80 km spans, giving 480 km of total fiber. A suitable source must offer stable wavelength, sufficient optical power, narrow spectral width, high modulation capability and compatibility with single-mode fiber.

Important transmitter characteristics

Parameter	Why it matters
Wavelength and grid	Each channel must remain aligned to its assigned DWDM wavelength; drift can cause adjacent-channel interference and filter loss.
Modulation speed	The source/modulator must support the 25 Gb/s data stream; the given approximation requires 17.5 GHz electrical/optical bandwidth.
Linewidth and side modes	Narrow linewidth and strong side-mode suppression reduce spectral overlap, phase noise and unwanted channel energy.
Output power and efficiency	Adequate launch power supports the receiver after span losses, while excessive power increases Kerr and scattering nonlinearities.
Chirp and dispersion	Frequency chirp interacting with fiber dispersion broadens pulses and closes the eye, especially over long distances.
Standards and safety	Optical interfaces must be checked against the relevant physical-layer specifications; accessible emission must be classified under the applicable laser-safety standard.

Standards context

ITU-T G.957 specifies optical-interface characteristics for certain SDH systems; it is useful as an interface-reference framework but does not by itself define this exact 8 × 25 Gb/s DWDM design. IEEE 802.3bs and IEEE 802.3bm cover particular high-speed Ethernet physical-layer interfaces, not a blanket approval for every DWDM transmitter configuration. The selected transmitter must be checked against the exact PHY/application being implemented, including wavelength, optical power, transmitter eye, extinction ratio, jitter and receiver compatibility. ANSI Z136 is a laser-safety framework; classification depends on wavelength, accessible emission, beam geometry, exposure duration and measurement conditions.

Problem Requirement Statement: Design an eight-channel 1550 nm DWDM transmitter with 100 GHz spacing and 25 Gb/s NRZ per channel, providing stable wavelengths within ±0.1 nm, adequate output power, at least 17.5 GHz modulation bandwidth, low chirp and spectral impurity, controlled nonlinear penalties, safe accessible emission, and a practical integration/DSP strategy for six 80 km amplified spans.

Stage 2 – Investigate and Select the Optical Source

Carrier recombination in a semiconductor active region can be radiative, producing photons, or non-radiative, converting carrier energy mainly into heat. The bandgap determines the approximate photon energy and emission wavelength. For the given bandgap:

$\lambda = 1.24 / E_g = 1.24 / 0.80 =$ **1.550 µm = 1550 nm**.

Internal quantum efficiency: $\eta_{int} = \tau_{nr} / (\tau_r + \tau_{nr}) = 40 / (10 + 40) =$ **0.80 (80%)**.

Optical-source comparison

Source	Strengths	Limitations for this link	Assessment
LED	Low cost, broad spectrum, simple drive	Low radiance, broad spectrum and limited high-speed suitability	Not suitable
FP laser	Higher power and speed than LED; relatively low cost	Multiple longitudinal modes and wider spectrum	Possible low-cost alternative, not preferred
DBR laser	Wavelength-selective grating; tunable/stable designs possible	More complex fabrication/control than basic DFB	Credible alternative where tunability is important
DFB laser	Single dominant longitudinal mode, narrow linewidth, good power and speed	Needs temperature/current control; direct modulation can produce chirp	Recommended source
VCSEL	Low power consumption, compact, array-friendly	Many conventional devices are optimized for shorter wavelengths and shorter-reach links; 1550 nm versions require specific qualification	Not default choice
QD laser	Potential low threshold, temperature tolerance and broad gain	Device maturity, availability and exact 1550 nm performance must be verified	Emerging alternative

Laser oscillation begins when round-trip modal gain equals or exceeds total cavity loss. Above threshold, stimulated emission dominates and coherent output builds. Linewidth describes the spectral width of the emitted carrier; side-mode suppression ratio (SMSR) compares the main mode with unwanted longitudinal modes. A narrow linewidth and high SMSR reduce interference with neighbouring DWDM channels and improve filtering and coherent/DSP performance.

Design Decision 1: I recommend a **1550 nm DFB laser** because it offers single-mode operation, narrow linewidth, high side-mode suppression, adequate output power and modulation capability suitable for DWDM. A DBR laser is a credible alternative when wide wavelength tuning or reconfigurable channel assignment is a stronger requirement than simplicity and cost.

Stage 3 – Modulation, Chirp and Spectral Stability

Output power calculation

$P = \eta_s(I - I_{th}) = 0.3$ mW/mA × (40 – 12) mA = **8.4 mW**.

$P(dBm) = 10 \log_{10}(8.4 \text{ mW} / 1 \text{ mW}) =$ **9.24 dBm**.

Bandwidth calculation

$B \approx 0.7 R_b = 0.7 \times 25$ Gb/s = **17.5 GHz**. The stated 3 dB modulation bandwidth is 20 GHz, which exceeds the estimated requirement by **2.5 GHz**. The nominal margin is about 14.3% of the required bandwidth. This is sufficient by the simplified relation, but final qualification must include driver, modulator, packaging and receiver bandwidth.

Carrier and photon rate equations

A common rate-equation model for a laser diode is:

$$dN/dt = I/(qV) - N/\tau_n - \Gamma v_g g(N,S)S$$

$$dS/dt = \Gamma\beta N/\tau_{sp} + \Gamma v_g g(N,S)S - S/\tau_p$$

Here N is carrier density, S is photon density, I is injection current, q is electron charge, V is active-region volume, Γ is optical confinement factor, v_g is group velocity, g is material gain, β is spontaneous-emission coupling factor, τ_n is carrier lifetime and τ_p is photon lifetime. Relaxation oscillation is the coupled carrier-photon response that produces a resonance

in the small-signal modulation response. Small-signal modulation assumes perturbations around a bias point; large-signal modulation involves substantial carrier and photon excursions, nonlinear gain, turn-on delay and possible transient chirp/overshoot.

Direct versus external modulation

Method	Advantages	Disadvantages	Decision
Direct modulation	Simple, compact and low cost; current directly controls optical output	Current-induced chirp and relaxation dynamics can interact with chromatic dispersion over 480 km	Not preferred for this multi-span DWDM link
External modulation	Laser can operate at steady bias; lower chirp with a suitable modulator; supports high-speed data	Adds modulator insertion loss, drive electronics and alignment/control complexity	Recommended

Wavelength drift and TEC

Uncontrolled drift = 0.1 nm/°C × 40 °C = **4.0 nm**. The channel spacing is approximately 0.8 nm, so the drift is 5.0 times the spacing. For a ±0.1 nm wavelength tolerance, the corresponding temperature variation is ±0.1/0.1 = **±1.0 °C** around the calibrated operating point, assuming the stated linear coefficient.

A thermoelectric cooler (TEC) with temperature sensor and feedback controller is recommended to hold the laser temperature within approximately ±1 °C. Injection-current tuning can provide fine wavelength adjustment, but it also changes carrier density, output power and chirp. TEC control is the primary stabilizer; current control may be used for limited fine tuning under calibrated feedback.

Design Decision 2: I recommend **external modulation and TEC-based wavelength control** because the link spans 480 km and has tight 0.1 nm wavelength tolerance. External modulation reduces chirp-related dispersion penalties, while the TEC prevents the calculated 4 nm uncontrolled drift.

Stage 4 – Nonlinear Impairments and Control

Effective length

Convert attenuation: $\alpha = 0.2$ dB/km ÷ 4.343 = 0.04605 km^{-1}. Then:

$L_{eff} = [1 - \exp(-\alpha L)]/\alpha = [1 - \exp(-0.04605 \times 80)]/0.04605 =$ **21.17 km**.

Self-phase modulation (SPM)

$\phi_{NL} = \gamma P L_{eff} = 1.3$ W^{-1}km$^{-1} \times 0.002$ W $\times$ 21.17 km = **0.0550 rad per span**. For six equally re-amplified spans, $\phi_{total} \approx 6 \times 0.0550 =$ **0.3302 rad**. This is a simplified accumulated phase estimate; actual phase depends on amplifier gain, span loss, power evolution and channel interactions.

XPM and FWM

Cross-phase modulation (XPM) occurs when the intensity of one WDM channel changes the refractive index experienced by another channel through the Kerr effect. It creates phase modulation and can convert to amplitude distortion in dispersive fiber. Its impact increases with channel power and channel overlap; dispersion and walk-off can reduce the interaction length. Four-wave mixing (FWM) combines frequency components to generate new frequencies such as f■ + f■ – f■. Products can fall on or near occupied DWDM channels, producing crosstalk. FWM is especially troublesome when channel spacing is uniform and phase matching is strong; fiber dispersion generally reduces phase matching.

SBS and SRS threshold calculations

Use $A_{eff} = 80$ µm^2 = 80 × 10^{-12} m^2 and $L_{eff} = 21.1$ km = 21,100 m in the supplied relations:

$P_{th,SBS} \approx 21\, A_{eff}/(g_B L_{eff}) = 21 \times 80\times10^{-12} / (5\times10^{-11} \times 21{,}100) =$ **1.59 mW**.

$P_{th,SRS} \approx 16\, A_{eff}/(g_R L_{eff}) = 16 \times 80\times10^{-12} / (6\times10^{-1}■ \times 21{,}100) =$ **1.01 W**.

Quantity	Calculated value	Comparison
SBS threshold (per supplied formula)	1.59 mW	Below 2 mW per channel; SBS may be relevant under the narrow-linewidth assumption.
SRS threshold	1.01 W	Far above 2 mW per channel and 16 mW aggregate launch power; SRS is not expected to reach threshold under these estimates.
Per-channel launch	2 mW	Eight channels give 16 mW aggregate launch power.

SBS is stimulated scattering from acoustic density waves and is usually strongly directional/backward, with a relatively narrow gain bandwidth. SRS is stimulated scattering from optical phonons, has a broader gain bandwidth and transfers power from higher to lower optical frequencies. SBS can reflect power toward the transmitter and limit usable launch power; SRS can redistribute power among channels. SBS threshold generally increases when the source linewidth is broadened beyond the Brillouin gain bandwidth, although broadening must not compromise DWDM spectral occupancy or adjacent-channel spacing.

Nonlinearity and safety strategy

Two practical nonlinear-penalty controls are: (1) optimize launch power per channel rather than maximizing it, balancing receiver OSNR against Kerr effects; and (2) use suitable dispersion management/channel planning and DSP compensation, including XPM/FWM-aware design where justified. Additional measures include keeping the optical spectrum clean, controlling amplifier gain tilt and avoiding unnecessary power peaks.

For eye safety, the aggregate transmitter launch is 8 × 2 mW = 16 mW. As a screening assumption only, if a project adopts a 10 mW accessible-emission threshold, 16 mW would exceed that assumed value. This is not a formal ANSI Z136 classification: the actual permissible exposure and class depend on accessible emission, wavelength, beam diameter/divergence, viewing geometry and exposure time. Use an enclosed optical path, interlocked access, warning labels, connector safety procedures and a documented laser-safety assessment.

Design Decision 3: I recommend a **controlled-launch-power strategy with channel-aware dispersion management, amplifier equalization and receiver DSP**. The SBS estimate is below the stated per-channel launch level and must be investigated; SRS is well below threshold risk at this power. Avoid simply increasing launch power to overcome span loss. Apply a formal laser-safety classification to the completed hardware.

Stage 5 – Soliton, Digital Backpropagation and Integration

Soliton condition and calculations

A fundamental optical soliton is formed when anomalous group-velocity dispersion balances Kerr self-phase modulation. For the given pulse parameter T■ = 15 ps and |β■| = 20 ps²/km:

$L_D = T■^2/|β■| = (15\ ps)^2/(20\ ps^2/km) =$ **11.25 km**.

For a fundamental soliton, $L_{NL} = L_D$, so $P■ = 1/(\gamma L_D) = 1/(1.3 \times 11.25) =$ **68.4 mW**.

At the stated 2 mW launch power: $L_{NL} = 1/(\gamma P■) = 1/(1.3 \times 0.002) =$ **384.6 km**.

$N^2 = L_D/L_{NL} = 11.25/384.6 = 0.0293$; N = **0.171**.

A fundamental soliton requires N = 1, which would require approximately 68.4 mW peak power under the given model. The stated 2 mW gives N far below one, so the specified operating point does not support fundamental-soliton transmission. Increasing power to the soliton value is not automatically acceptable because it would aggravate nonlinear penalties and the SBS concern. Conventional dispersion compensation and DSP are more appropriate.

Digital backpropagation (DBP)

DBP numerically reverses the propagation model of the fiber, approximating the inverse of dispersion and Kerr nonlinear propagation. It can compensate accumulated chromatic dispersion and some SPM/XPM effects when the channel waveform and link parameters are sufficiently known. Practical limitations include computational complexity, sensitivity to model errors and amplifier noise, and the need for access to the sampled waveform. In a multi-channel system, single-channel DBP cannot fully remove inter-channel nonlinearities; multi-channel DBP is more computationally demanding. For this 8-channel link, use receiver DSP for chromatic dispersion and residual intrachannel nonlinearities, and consider multi-channel DBP only if measured performance justifies the complexity.

Standards verification

Check the exact optical interface/PHY selected for deployment. Relevant measurements include center wavelength and wavelength tolerance, average launch power, extinction ratio, transmitter eye mask, spectral characteristics, jitter, optical return loss, receiver sensitivity and overload, and channel spacing/filter compatibility. ITU-T G.957 is relevant to specified SDH optical interfaces; IEEE 802.3bs/bm are Ethernet PHY standards for their defined applications. The proposed 8 × 25 Gb/s DWDM configuration must be mapped to an applicable standardized interface or treated as a system-specific design requiring its own interoperability validation.

Integration alternatives

Implementation	Advantages	Challenges	Assessment
Discrete components	Flexible component selection, easier individual replacement and characterization	Larger footprint, more fiber connections, alignment and packaging overhead	Good for prototype, less scalable
Monolithic InP	Laser, modulator and other photonic functions can be integrated on one platform	Process complexity, yield and thermal/electrical design constraints	Strong option for compact optical engine
Si-photonics + hybrid InP laser	Scalable passive routing, muxing and integration with electronics; laser supplied by hybrid III-V source	Hybrid coupling, thermal control, packaging and reliability qualification	Recommended platform for scalable multi-channel module

Design Decision 4: I recommend a **silicon-photonics transmitter with hybrid-integrated InP DFB lasers**, external modulators and integrated wavelength multiplexing. It offers a compact and scalable multi-channel implementation while retaining a suitable III-V gain source. Include TEC feedback, per-channel power monitoring and receiver-side DSP. A discrete implementation is appropriate for early prototyping or low-volume deployment; monolithic InP is a credible alternative where a highly integrated optical engine is available and qualified.

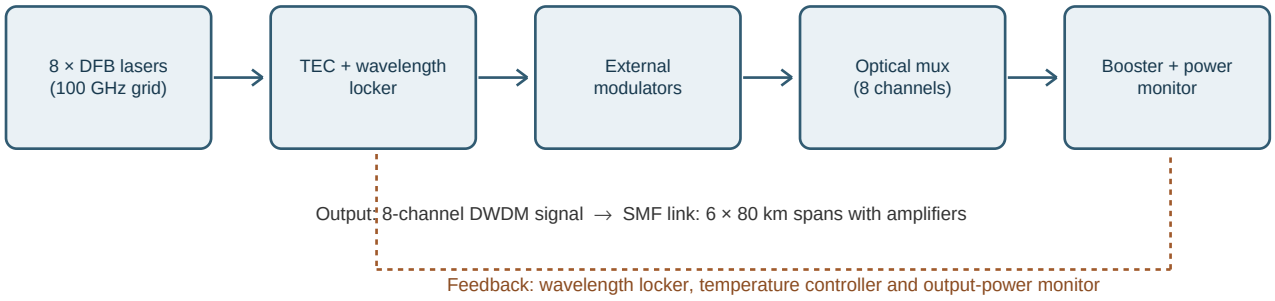
Stage 6 – Verification and Critical Evaluation

Requirement	Target / calculated result	Achieved?	Evaluation
Optical source	1550 nm DFB laser, single dominant mode	Yes, design selection	Appropriate for narrow-grid DWDM; verify actual linewidth and SMSR from datasheet/test.
Output power	8.4 mW = 9.24 dBm at 40 mA	Yes, calculation	Meets the stated slope-efficiency model; actual coupled launch power must include losses.
Modulation bandwidth	17.5 GHz required; 20 GHz available; 2.5 GHz margin	Yes, nominally	Adequate by given approximation; validate full electro-optic link response.
Wavelength stability	±0.1 nm; TEC stability approximately ±1 °C	With TEC	Uncontrolled 4 nm drift is unacceptable; feedback control is mandatory.
Nonlinear effects	SPM 0.055 rad/span; SBS estimate 1.59 mW; SRS 1.01 W	Mitigation required	SBS estimate is below launch power; validate threshold with actual linewidth and fiber parameters.
Soliton condition	L_D=11.25 km; N=0.171	No soliton	N is not close to 1; use dispersion compensation/DSP instead.
Integration / DSP	Si-photonics with hybrid InP DFB; receiver DSP	Recommended	Scalable architecture; DBP complexity and standards mapping require system validation.
Eye safety	16 mW aggregate transmitter launch	Requires assessment	Cannot assign ANSI class from power alone; assess accessible emission and geometry.

The calculations are broadly consistent with the original communication requirement if the design uses external modulation, active wavelength stabilization and controlled launch power. The source power and nominal bandwidth are adequate in the simplified model. The uncontrolled wavelength drift is not acceptable. The SBS estimate identifies a potential issue at the proposed launch power and requires practical validation, including linewidth broadening and actual span conditions. Soliton transmission is not feasible at the stated 2 mW peak-power assumption; dispersion compensation and DSP are the appropriate route. Standards and eye-safety compliance cannot be claimed from these calculations alone and must be demonstrated by testing.

Final Transmitter Design – Integrated Engineering Solution

Proposed block diagram



Final design parameter table

Parameter	Proposed value / method	Reason
Optical source	Eight 1550 nm DFB laser channels	Narrow linewidth and single-mode spectral output
Channel spacing	100 GHz (approximately 0.8 nm)	Matches assignment DWDM grid
Laser bias	40 mA; threshold 12 mA	Given operating point
Calculated source output	8.4 mW / 9.24 dBm per laser (model result)	From slope efficiency relation
Modulation	External modulator; 25 Gb/s NRZ	Reduces current-induced chirp

Parameter	Proposed value / method	Reason
Bandwidth	20 GHz available; 17.5 GHz estimated requirement	2.5 GHz nominal margin
Wavelength control	TEC feedback; approximately ±1 °C control around setpoint	Maintains ±0.1 nm wavelength tolerance under stated coefficient
Multiplexing	Integrated 8-channel optical mux	Combines channels on 100 GHz grid
Power management	Per-channel power monitor and controlled booster	Balances receiver power and nonlinear penalty
Fiber link	6 × 80 km amplified SMF spans	480 km total span distance
Compensation	Receiver dispersion equalization and DSP; evaluate multi-channel DBP if needed	Avoids impractical soliton power and mitigates residual impairments
Integration	Si-photonics with hybrid InP DFB laser array	Compact and scalable multi-channel platform

Final Engineering Decision: YES – WITH CONDITIONS

The proposed architecture is suitable for further engineering development and controlled deployment after qualification. The DFB source, external modulation, TEC wavelength locking, integrated mux and DSP form a coherent solution for the 8-channel DWDM requirement. Deployment is conditional on: (1) measuring actual laser linewidth, SMSR, output power and wavelength stability; (2) validating SBS threshold and nonlinear performance on the selected fiber at the real amplifier launch powers; (3) verifying the exact applicable IEEE/ITU interface requirements and interoperability; (4) completing ANSI Z136 laser-safety classification for accessible emission; and (5) testing BER, eye diagrams, OSNR and performance over all six spans. The design should not be described as standards-compliant or eye-safe until these tests are passed.

Conclusion

The engineering study selects a TEC-stabilized DFB laser source with external modulation for an 8-channel, 25 Gb/s DWDM transmitter near 1550 nm. The calculations show 1550 nm emission, 80% internal quantum efficiency, 8.4 mW output at the stated bias, and a 20 GHz modulation bandwidth against a 17.5 GHz estimated requirement. Temperature control is essential because the uncontrolled wavelength drift is 4 nm. Nonlinear analysis indicates a need to investigate SBS, control channel launch power and manage dispersion; the SRS threshold is much higher than the stated launch powers. The soliton order is much less than one, so soliton transmission is not recommended. A silicon-photonics platform with hybrid InP DFB lasers, external modulators and receiver DSP is the preferred scalable implementation. Final deployment requires standards, eye-safety and end-to-end performance verification.

References and Standards for Verification

1. Assignment brief: ECA18 – Optoelectronics and Optical Communication, CO2 Problem-Solving Assignment (provided source document).

2. ITU-T Recommendation G.957, Optical interfaces for equipment and systems relating to the synchronous digital hierarchy (SDH).

3. IEEE 802.3bs and IEEE 802.3bm Ethernet physical-layer specifications; consult the exact PHY clauses applicable to the selected system.

4. ANSI Z136 series, Safe Use of Lasers; apply the current relevant standard and institutional laser-safety procedures.

5. Numerical relations and device parameters used in this report are those supplied in the assignment brief. Assumptions are stated alongside the calculations.